

BASAL GANGLIA & ITS FUNCTIONS

Dr. Fatima Haq

Lecturer

DIKIOHS, DUHS.

LEARNING OBJECTIVES

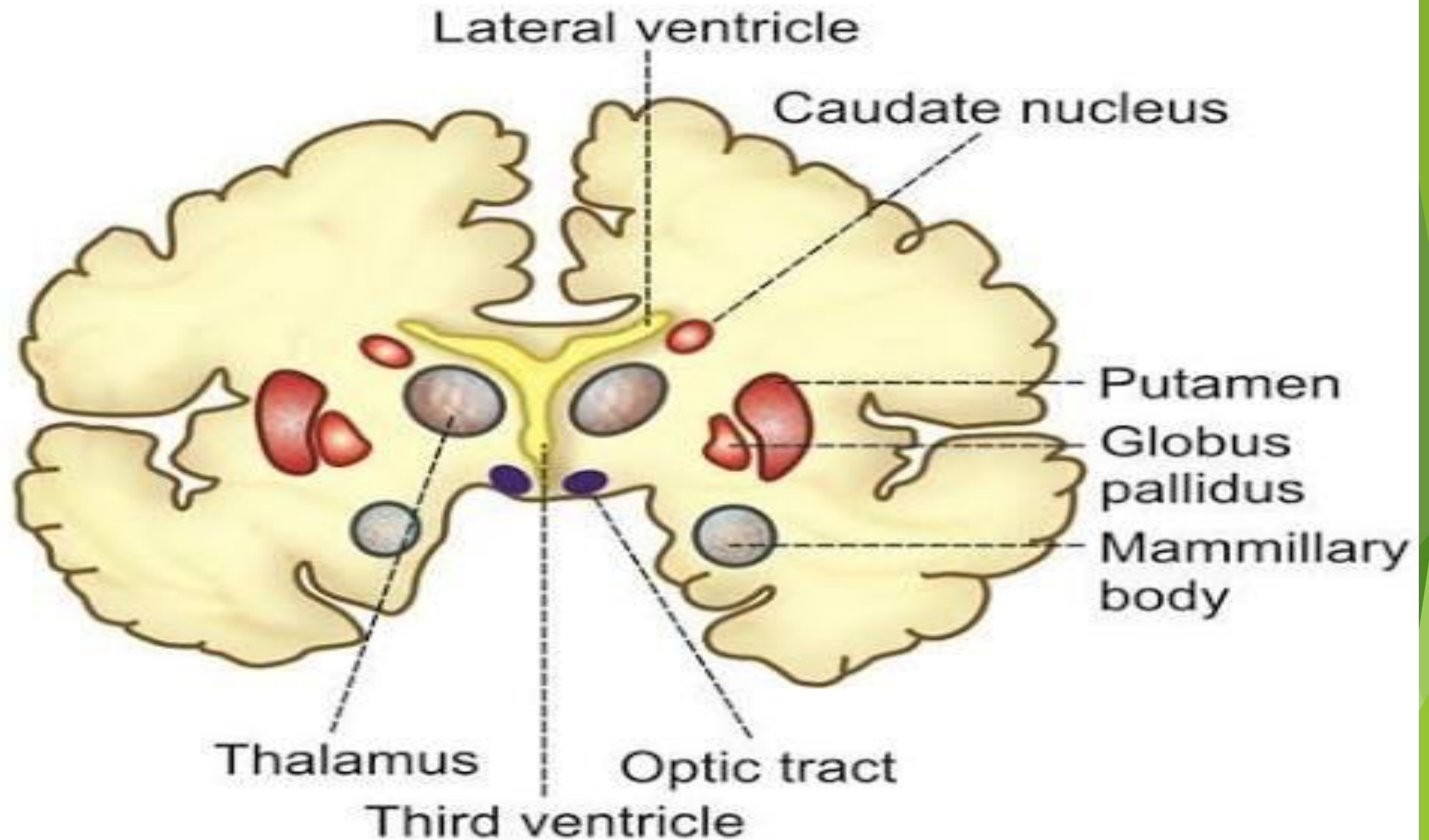
By the end of this lecture, students should be able to:

- ✓ Define basal ganglia & describe its major components.
- ✓ Identify the anatomical location & basic organization of basal ganglia.
- ✓ Describe input/output connections & neuronal circuits of basal ganglia.
- ✓ Describe the functions of basal ganglia.
- ✓ Correlate abnormalities of basal ganglia with common movement disorders like parkinsonism & Huntington disease.

BASAL GANGLIA

Basal ganglia are the scattered masses of gray matter located within the cerebral hemispheres.

- These ganglia are located lateral to & surrounding the thalamus, occupying large portions of interior regions of both cerebral hemispheres.
- Basal ganglia form the part of extrapyramidal system & is concerned with motor activities.
- Almost all motor & sensory nerve fibers connecting the cerebral cortex & spinal cord pass through the space '***the internal capsule of brain***' that lies between the major masses of basal ganglia(i.e. the caudate nucleus & putamen)
- Like cerebellum, basal ganglia constitute another **accessory motor system** that do not directly initiate voluntary movement but they help the brain to select appropriate movements, suppress unwanted movements & make movements smooth & coordinated.
- They also regulate the intensity & timing of movements.

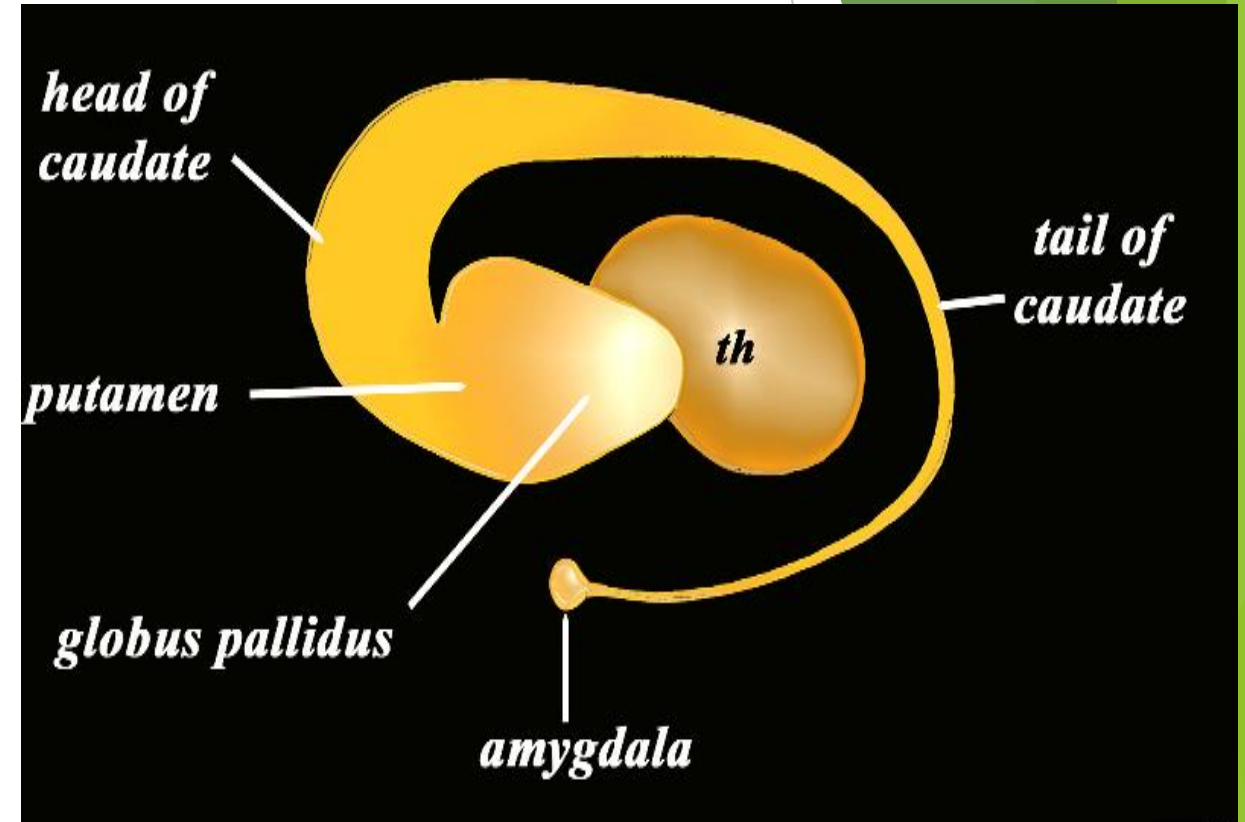


COMPONENTS OF BASAL GANGLIA

On each side of brain, these ganglia consist of:

► CAUDATE NUCLEUS

- It is a long, curved nucleus closely related to lateral ventricle.
- Caudate nucleus has a *head portion* situated rostral to thalamus & a *tail portion* that extends along the dorsolateral surface of thalamus.
- The tail of caudate nucleus ends in *amygdaloid nucleus*.



► PUTAMEN

- It is located lateral to the internal capsule of brain.

► GLOBUS PALLIDUS

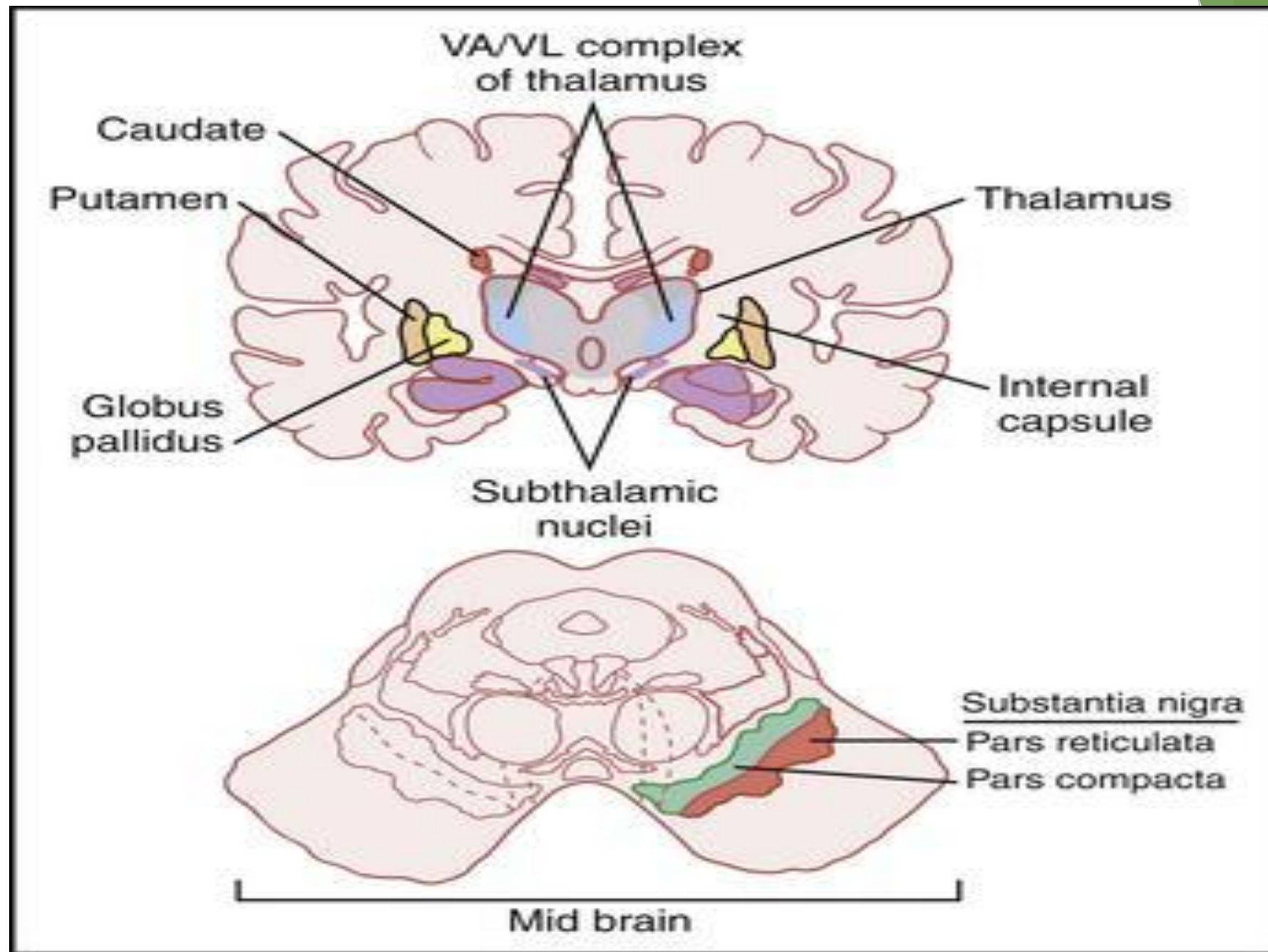
- It is divided into two segments, the external segment termed as '*GPe*' & internal segment '*GPi*'.

► SUBSTANTIA NIGRA

- It is located in the midbrain situated below the red nucleus.
- Substantia nigra is an important source of dopamine.

► SUBTHALAMIC NUCLEUS

- It is present lateral to red nucleus & dorsal to substantia nigra.

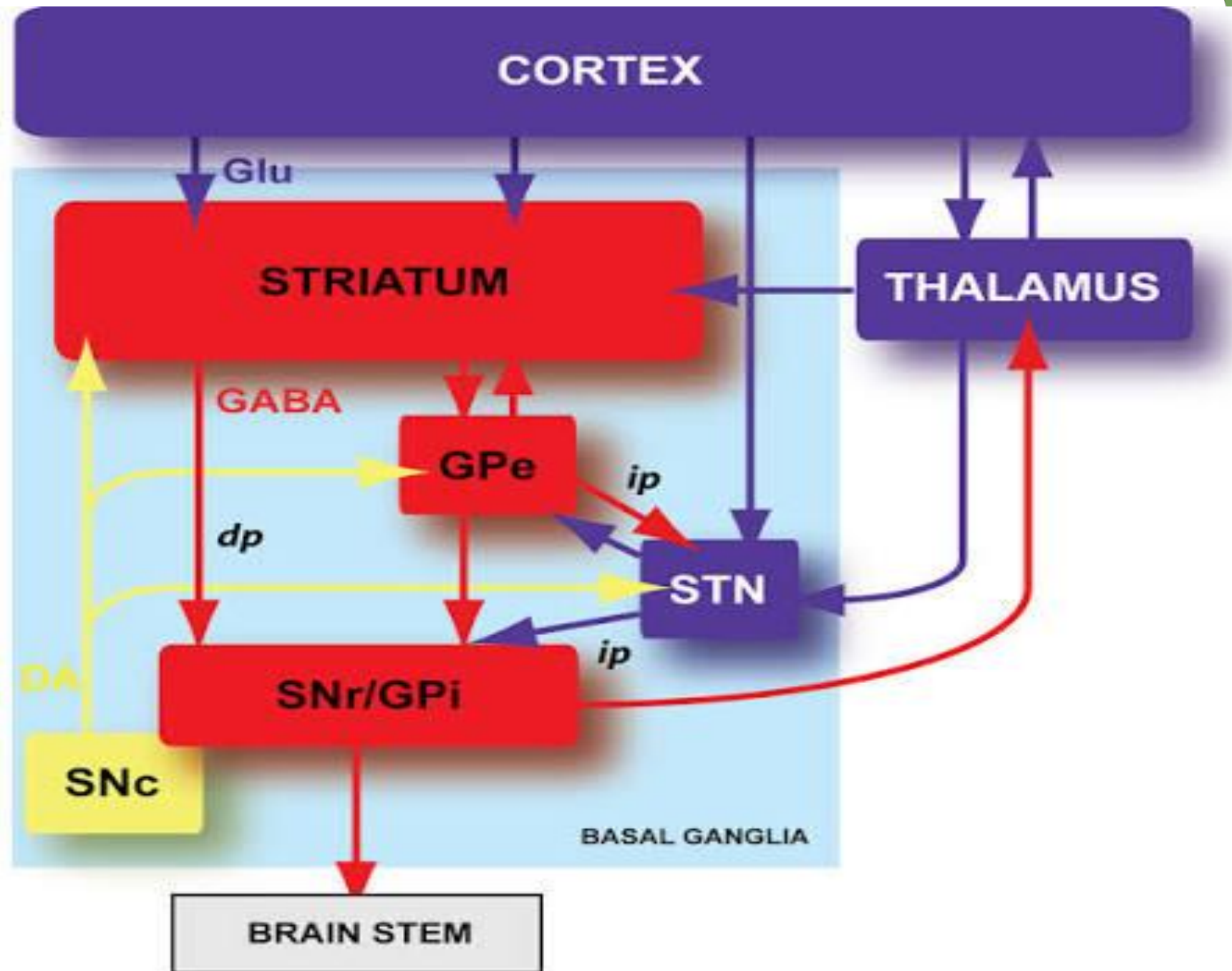


STRIATUM

- It is an important functional part of basal ganglia & consists mainly of caudate nucleus & putamen.
- Striatum receives a large amount of information from the cerebral cortex & acts as an important *input station* of basal ganglia.

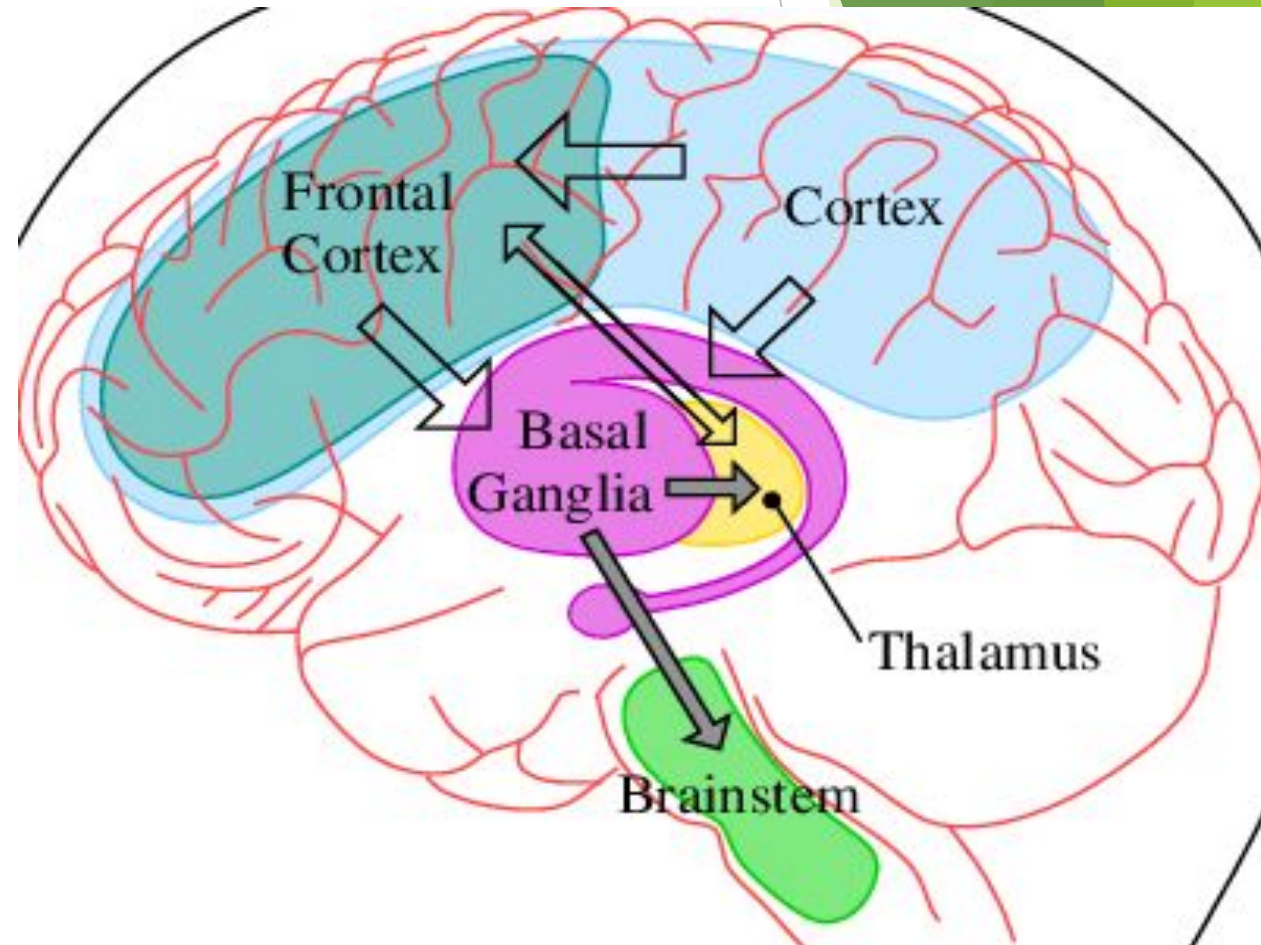
LENTIFORM NUCLEUS

- Lentiform nucleus consists of putamen & globus pallidus.
- GPi/SNr acts as major *output station* of basal ganglia.



CORTICO-BASAL GANGLIA-THALAMO-CORTICAL LOOP

- It is a system of neuronal circuits in the brain & involves connections between the cortex, basal ganglia, thalamus & back to the cortex.
- There are two main pathways inside this loop:
 - ▶ **DIRECT PATHWAY** speeds up or lets movements happen.
 - ▶ **INDIRECT PATHWAY** slows down or stops unwanted movements.



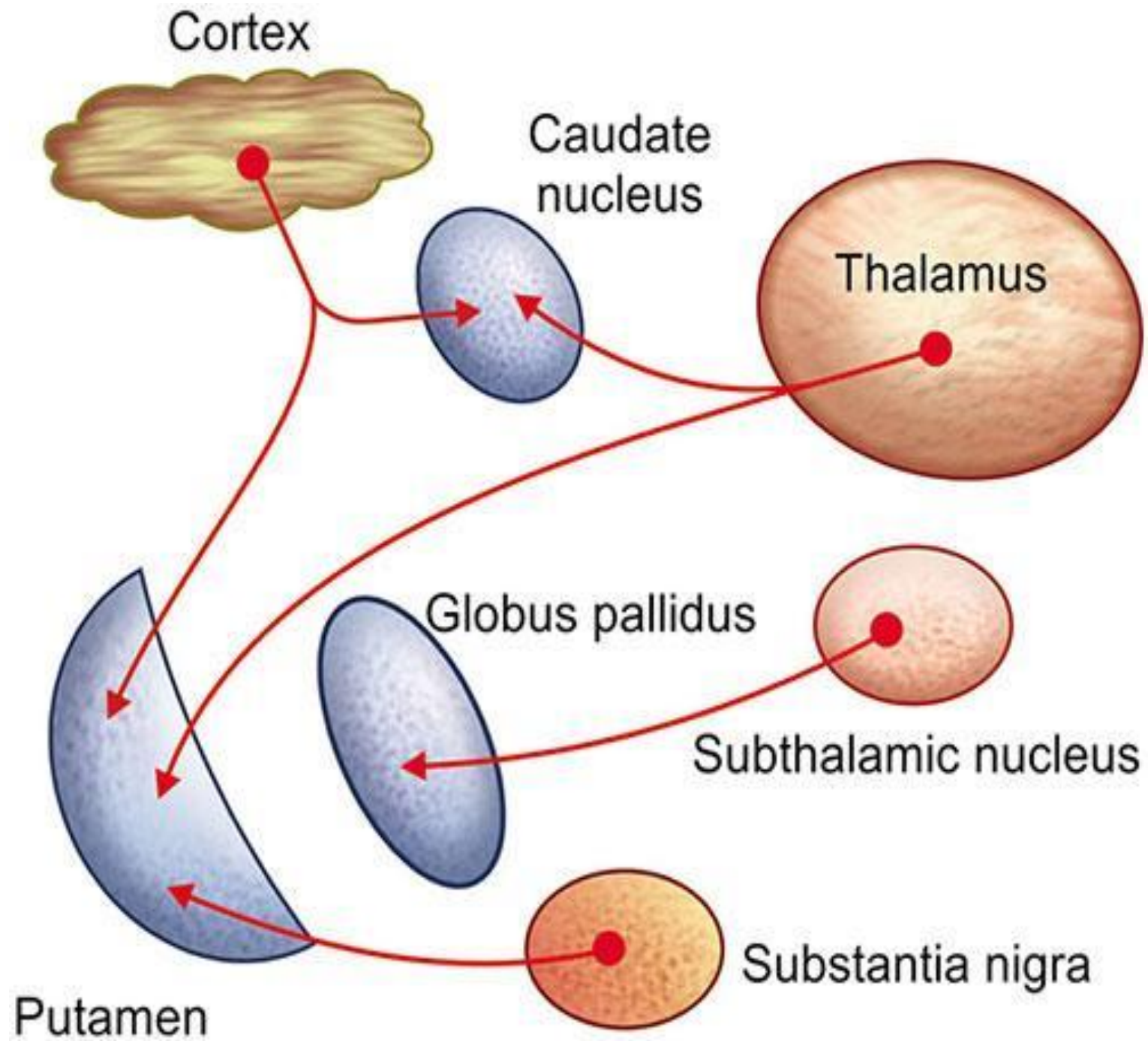
CONNECTIONS OF BASAL GANGLIA

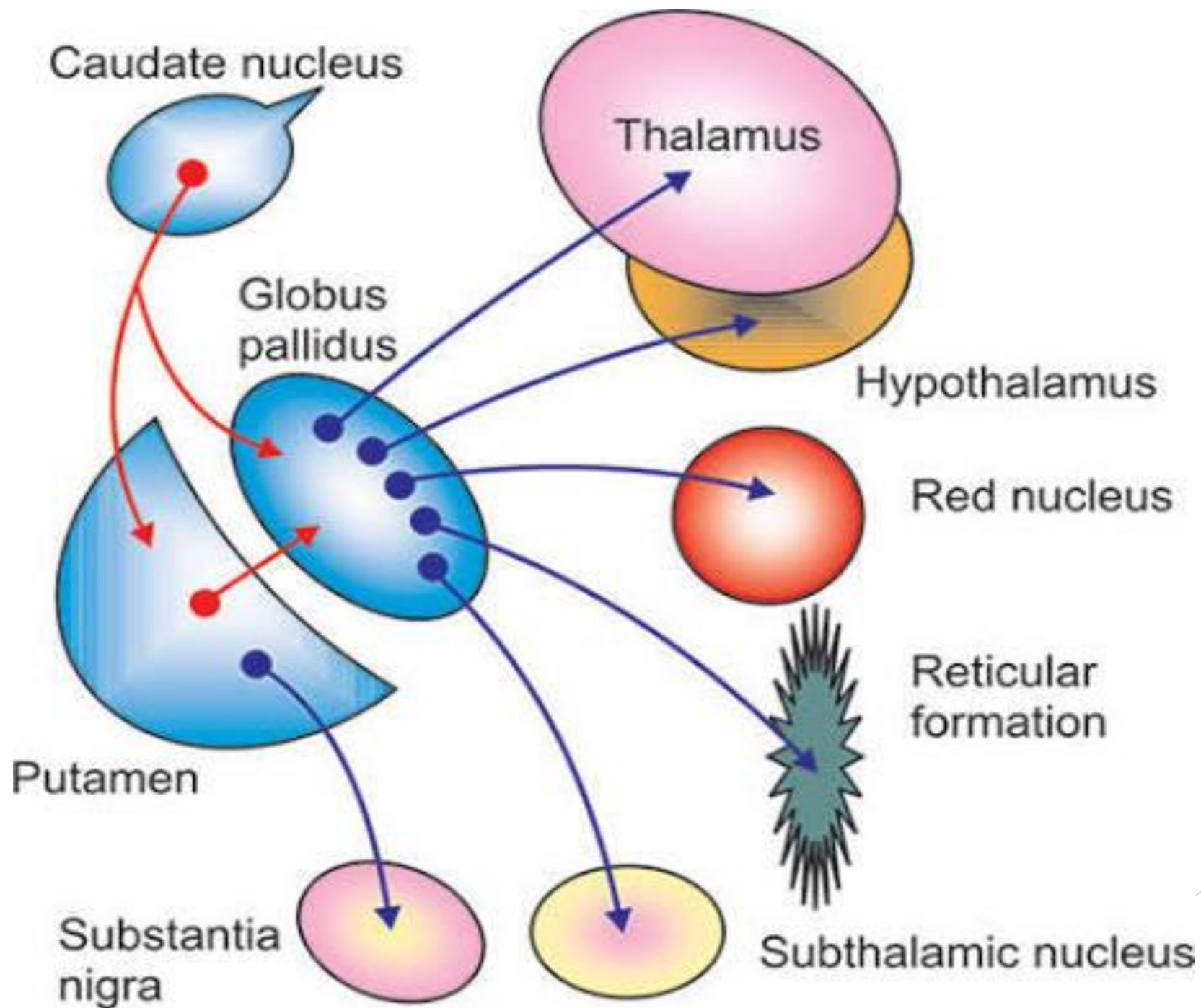
INPUT OR AFFERENT CONNECTIONS

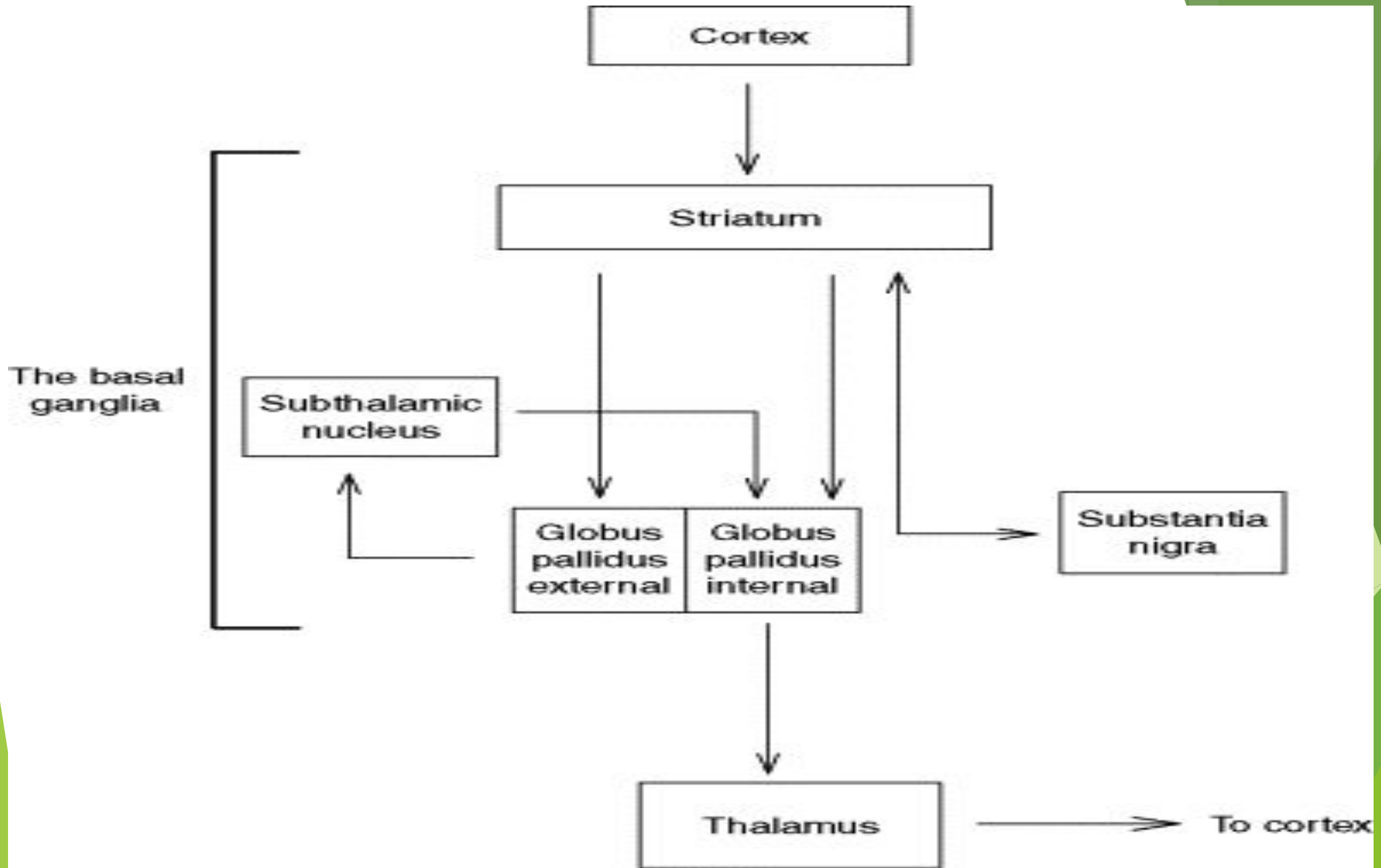
- ▶ **Corticostriate pathway:** send signals from cortex into the striatum.
- ▶ **Thalamostriate pathway:** send signals from thalamus into the striatum.

OUTPUT OR EFFERENT CONNECTIONS

- ▶ **Pallidothalamic & nigrothalamic pathways:** send final signals out of basal ganglia to the thalamus & brainstem which then report back to the motor cortex.







FUNCTIONS OF BASAL GANGLIA

1. CONTROL OF MUSCLE TONE

- ❑ Gamma motor neurons of spinal cord are responsible for the development of tone in the muscles.
- ❑ Basal ganglia control muscle tone by inhibiting gamma motor neurons through descending inhibitory system in brainstem.
- ❑ Lesions of basal ganglia increase muscle tone leading to muscle rigidity.

2. REGULATION OF VOLUNTARY MOVEMENTS

- ❑ Movements of voluntary motor activity are initiated by cerebral cortex, however controlled by basal ganglia which are in close association with cerebral cortex.
- ❑ Basal ganglia control these motor activities by the role of neuronal circuits between basal ganglia & other parts of brain involved in voluntary motor movements(i.e. the cerebral cortex).

- Nerve fibers from cerebral cortex reach the caudate nucleus.
- From here, the fibers go to putamen however some fibers from cerebral cortex go directly to putamen also.
- Fibers from putamen go to globus pallidus & then run towards the thalamus, subthalamic nucleus & substantia nigra(subthalamic nucleus & substantia nigra are in turn projected into the thalamus).
- Fibers from thalamus are then projected back into the cerebral cortex.

3. REGULATION OF CONSCIOUS MOVEMENTS

- Fibers between cerebral cortex & caudate nucleus are associated with the regulation of conscious movements known as ***cognitive control of activity***.

4. REGULATION OF SUBCONSCIOUS MOVEMENTS

- Cortical fibers reaching the putamen are associated with the regulation of subconscious movements that take place during skilled motor activities (e.g. writing a learnt alphabet or cutting paper).

5. CONTROL OF REFLEX MUSCULAR ACTIVITY

- Some reflex muscular activities are important in maintaining posture (especially visual & labyrinthine reflexes).
- Basal ganglia are involved in the coordination & integration of impulses for these reflex activities.
- Lesions of basal ganglia result in abnormal reflex activities particularly visual & labyrinthine reflexes.

6. CONTROL OF AUTOMATIC ASSOCIATED MOVEMENTS

- Movements in the body that take place along with some motor activities are termed as automatic associated movements like swing of arms while walking, appropriate facial expressions while talking etc.
- Basal ganglia are responsible for these type of movements.
- Lesions of basal ganglia result in loss of automatic associated movements e.g. **mask-like face**(face without appropriate expressions while doing any work), **statue-like body**(body without associated movements).

7. ROLE IN AROUSAL MECHANISM

- Globus pallidus & red nucleus are involved in arousal mechanism.
- Lesions in these areas cause drowsiness leading to sleep.

Functions of Basal Ganglia

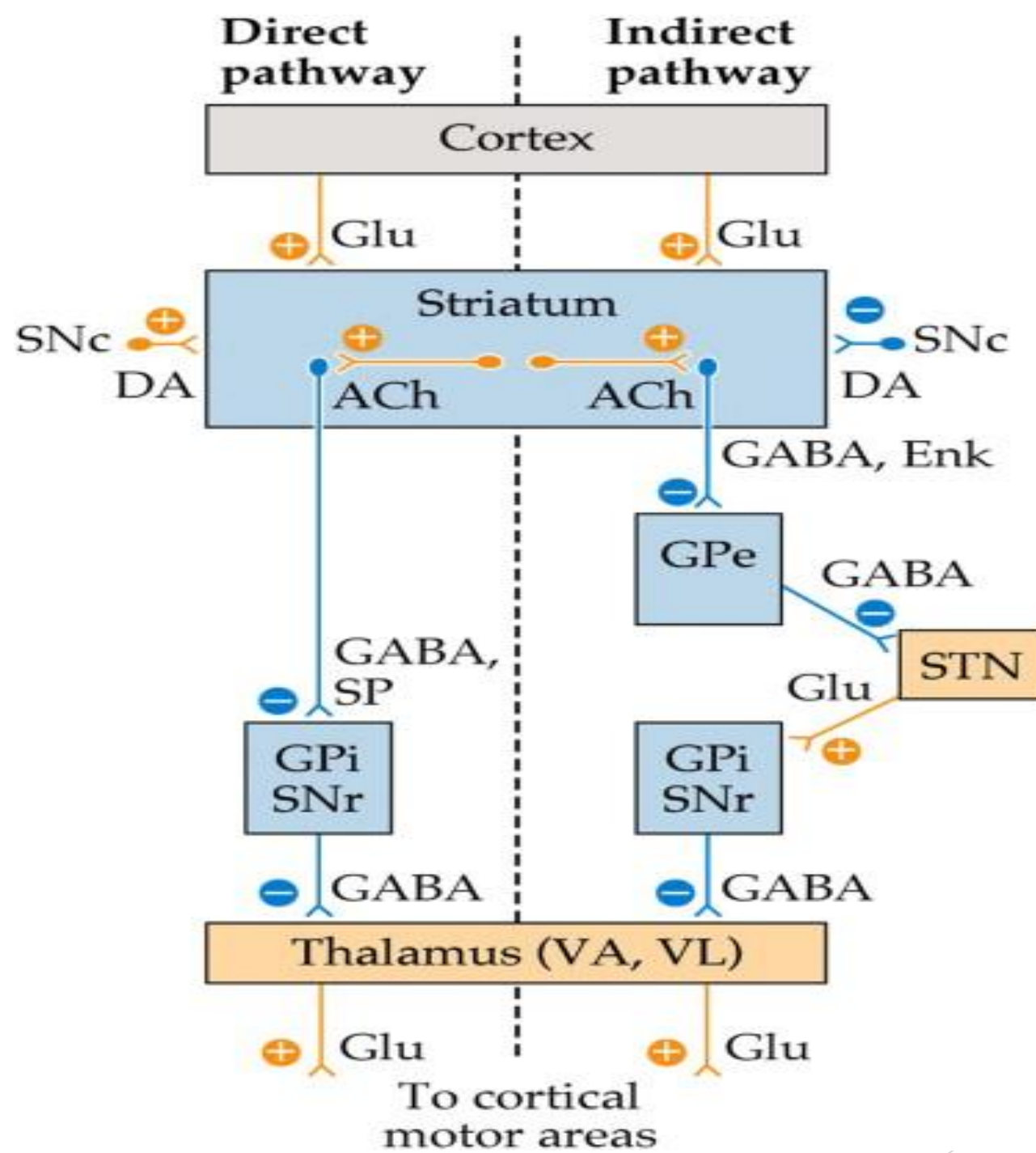
- Act by modifying ongoing activity in motor pathways
- Inhibit muscle tone (proper tone – balance the excitatory and inhibitory inputs to motor neurons that innervate skeletal muscle)
- Select and maintain purposeful motor activity while suppressing unwanted patterns of movement
(Action Selection: behavioral switching or decision making)
- Monitor and coordinate slow and sustained contractions, especially those related to posture and support
- Regulate attention and cognition
- Control timing and switching
- Motor planning and learning

NEUROTRANSMITTERS IN THE FUNCTIONING OF BASAL GANGLIA

- ▶ **DOPAMINE:** secreted by dopaminergic fibers from substantia nigra to putamen & caudate nucleus. Deficiency of dopamine causes *parkinsonism*.
- ▶ **GABA(Gamma-aminobutyric acid):** secreted by fibers of corpus striatum & substantia nigra.
- ▶ **ACETYLCHOLINE:** released by fibers from cerebral cortex to caudate nucleus & putamen.
- ▶ **SUBSTANCE P & ENKEPHALINS:** released by fibers from globus pallidus reaching substantia nigra.
- ▶ **NORADRENALINE:** secreted by fibers b/w basal ganglia & reticular formation.
- ▶ **GLUTAMIC ACID:** released by fibers from subthalamic nucleus to globus pallidus & substantia nigra.

Neurotransmitters of basal ganglia

1. Dopamine pathway: from substantia nigra to caudate nucleus and putamen.
2. Gama amino butyric acid pathway from caudate nucleus and putamen to globus pallidus and substantia nigra.
3. Acetylcholine pathway from cortex to the caudate nucleus to putamen.
4. Glutamate that provide the excitatory signals that balance out the large no. of the inhibitory signals transmitted specially by the dopamin, GABA & serotonin inhibitory transmitters.



APPLIED PHYSIOLOGY

1. PARKINSON DISEASE

- Slowly progressive degenerative disease of nervous system associated with the destruction of brain cells that produce dopamine.
- Causes include viral infection of brain e.g. encephalitis, cerebral arteriosclerosis, injury to basal ganglia or destruction or removal of dopamine in basal ganglia eg. long-term treatment with antihypertensive drugs.

SIGNS & SYMPTOMS

- Tremor occurs during rest but disappears while doing work.
- Movements start slowing down over the time(patient taking long time to perform simple task) & gradually becomes unable to initiate voluntary movements(***akinesia***) or voluntary movements are reduced(***hypokinesia***).
- Poverty of movements(***statue-like body, mask-like face***) due to absence of automatic associated movements.

- Rigidity of limbs resulting from muscles stiffness caused due to removal of inhibitory influence on gamma motor neurons.
- Rigidity extends to neck & trunk in later stages.
- Patient loses normal gait & walks quickly in short steps by bending forward(*festinant gait*).
- Patient develops speech problem(may speak very softly or sometimes rapidly) with words repeated many times & finally speech becomes slurred & patient hesitates to speak.
- In later stages of Parkinson disease, patient may develop dementia.

TREATMENT OF PARKINSON DISEASE

- Parkinson disease is treated with dopamine injection in the form of levodopa along with carbidopa to prevent excess dopamine content in blood & liver.
- Some symptoms of Parkinson disease such as tremor are abolished by surgical intervention.

2. WILSON DISEASE

- It is an inherited disorder characterized by *excessive copper in body tissues*.
- In Wilson disease, copper is deposited in liver, brain, kidneys & eyes causing tissue damage & eventually affected organs stop functioning.

3. CHOREA

- It is characterized by rapid jerky involuntary movements mostly involves the limbs & is due to lesion in caudate nucleus & putamen.

4. ATHETOSIS

- Athetosis is characterized by slow rhythmic twisting involuntary movements & is due to lesion in caudate nucleus & putamen.

5. HUNTINGTON CHOREA

- It is an inherited progressive neural disorder due to degeneration of neurons releasing GABA in corpus striatum & substantia nigra.
- Huntington chorea is characterized by chorea, hypotonia & dementia.

6. KERNICTERUS

- Kernicterus is the brain damage mainly affecting basal ganglia in infants due to severe jaundice.

7. HEMIBALLISMUS

- It is a disorder characterized by violent involuntary movements on one side of body involving mostly the arm.
- The arm swings widely while walking due to absence of inhibitory influence on movements(*flinging movements*).

POST ASSESSMENT

Q1. Which structure is considered as the main input nucleus of basal ganglia?

- a) Globus pallidus internus(GPi)
- b) Globus pallidus externus(GPe)
- c) Striatum
- d) Substantia nigra pars compacta(SNc)

Q2. The Globus pallidus internus(GPi) normally exerts its major effect on thalamus through:

- a) Excitation
- b) Inhibition
- c) Dopaminergic stimulation
- d) Cholinergic stimulation

Cont..

Q3. Which clinical feature is most characteristic of Parkinson disease?

- a) Flaccid paralysis
- b) Resting tremor
- c) Intention tremor
- d) Hemiballismus

Q4. Which neurotransmitter is released by neurons of substantia nigra pars compacta(SNc)?

- a) GABA
- b) Glutamate
- c) Acetylcholine
- d) Dopamine